Supply Chain & Operations Analytics - Delta Sky Broadcasting

Abstract

This report presents a data-driven analysis of key operational processes at Delta Sky Broadcasting, a UK-based satellite television service provider. The focus areas include forecasting subscription call demand, inventory management of satellite boxes, engineer job allocation, and vehicle servicing efficiency. Using Microsoft Excel for data visualisation, forecasting, simulation, and optimisation, the study applies practical analytical techniques to evaluate current performance and recommend improvements. Time series analysis revealed a strong weekly pattern in call volume, with demand peaking midweek. Among various forecasting methods tested, exponential smoothing using Excel's FORECAST.ETS function demonstrated the best performance due to its adaptability and accuracy. Inventory simulation over seven days helped calculate total and average daily costs, highlighting the trade-off between holding, ordering, and backorder penalties.

Linear programming was used to optimise engineer job assignments, reducing the total distance travelled from 28 to 27 miles compared to the existing manual approach. The queuing simulation for vehicle servicing, using Poisson arrivals and exponential service times, revealed moderate mechanic idle time and some vehicle delays, indicating room for improvement in scheduling. Based on these findings, the report recommends adjusting reorder points in inventory management, adopting LP-based job allocation to reduce operational travel, and implementing appointment-based servicing or additional mechanic support during peak demand. The report demonstrates how Excel-based analytical methods can guide more efficient resource management and enhance service delivery. The suggested improvements are cost-effective, scalable, and tailored to the operational needs of Delta Sky Broadcasting.

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Introduction

Delta Sky Broadcasting is a satellite television service provider based in the UK. The company wants to enhance how it handles new customer subscription requests, a significant element of its business. Currently, processes such as estimating demand, maintaining inventory, assigning engineers, and repairing cars are done manually or according to set protocols, which may be inefficient (Barrow, 2025).

As part of this project, an in-depth evaluation was performed on each operating area using Microsoft Excel. The study contains demand trend analysis, forecasts with different models, inventory cost simulation, task assignment optimisation with linear programming, and a vehicle service queue simulation. These strategies supplied useful information on how the company might enhance performance, cut expenses, and make better resource allocation decisions.

As a data analytics consultant, my purpose with this report is to use data-driven methodologies to provide practical recommendations. The organisation may boost productivity and customer happiness by spotting patterns in subscription demand, optimising scheduling and logistics, and enhancing inventory and maintenance methods. These suggestions are supported by quantitative data, guaranteeing they are feasible, quantifiable, and aligned with the company's operational objectives.

Investigating New Subscription Call Demand

Date	New Calls	CMA
01-May	1,576	
02-May	2,299	
03-May	1,878	
04-May	=AVERAGE(C5:C11)	
05-May	AVERAGE(number1, [number2], ...)	
06-May	1,240	1,622
07-May	864	1,638
08-May	1,882	1,674

Figure 1 Calculated CMA Value

A time series analysis was used on 56 days of data from 1st May to 25th June to understand the pattern of new subscription calls. First, a 7-day Centred Moving Average (CMA) was calculated (**Figure 1**) to smooth out short-term fluctuations while highlighting the underlying trend (Morgan, n.d.). The CMA was calculated by taking the average of 3 days before and after each point (e.g. for the 4th May, the average of 1st May to 7th May was used) (Barrow, 2025).

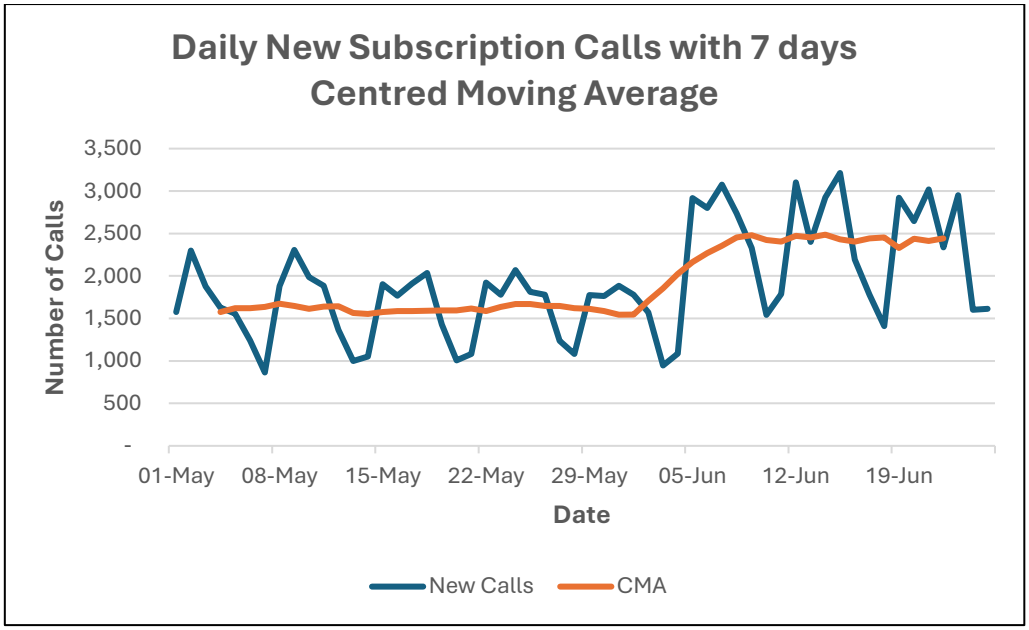

Figure 2 Line Graph of Daily New Subscription Calls

The Line chart **(Figure 2)** illustrates the New Calls in blue and CMA in Orange. The diagram shows a consistent pattern in May, but calls increased sharply in early June, indicating higher demand in the last few weeks.

Next, detrended daily values were calculated using the additive decomposition method because the seasonal effects stayed pretty constant over time and did not grow with the trend, making it more appropriate than a multiplicative model (Penn State University, 2024).

Detrended Value = Original Value − Trend (CMA) (Penn State University, 2024)

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	**Total**
Mon		208	329	335	162	755	630	590	**3,009**
Tue		659	182	144	181	531	- 49	209	**1,856**
Wed		375	324	402	340	720	440	608	**3,210**
Thu	51	246	445	144	235	282	781	- 104	**2,081**
Fri	- 66	- 278	- 165	132	- 135	- 153	- 212		**- 878**
Sat	- 382	- 568	- 587	- 410	- 907	- 885	- 661		**- 4,401**
Sun	- 774	- 503	- 536	- 538	- 940	- 617	- 1,047		**- 4,954**

Figure 3 Seasonal Matrix

After removing the CMA from the Original Values (New Calls), the remaining values show the seasonal and random patterns. These detrended values were arranged into a seasonal matrix grouped by day of the week and week number **(Figure 3)**. This helps identify repeating patterns on the same day of each week (Otexts, 2011).

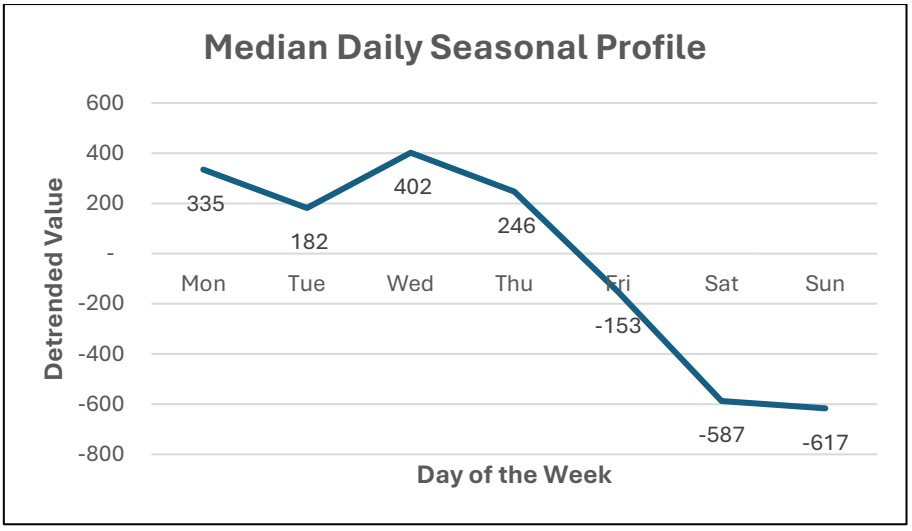

Figure 4 Line Chart of Median Daily Seasonal Profile

The daily seasonal profile **(Figure 4)** was created by taking the median detrended value from the seasonal matrix for each day of the week. This demonstrates that calls tend to be most significant midweek (especially Wednesday) and lowest on weekends, particularly Sunday. These results indicate substantial weekly seasonality in customer behaviour.

From this analysis, the organisation could consider boosting personnel and technical assistance on busy days such as Wednesdays while cutting staff on weekends. This would make it easier to manage resources and provide better customer service. It also implies that future forecasting algorithms should use daily seasonality to improve accuracy. Understanding this trend can also help you schedule marketing campaigns or promotions that match high-demand days. Aligning operational planning with customer behaviour may boost efficiency, save costs, and increase customer happiness.

Forecasting New Subscription Call Demand

Three seasonal models were used to predict daily call demand: Seasonal Naïve, Seasonal Average, and Seasonal Exponential Smoothing. The data from May 1 to June 18 was utilised for training, and a 7-day forecast from 19th to 25th June was created for each method to test how well the models predicted future calls. For each approach, one-step-ahead forecasts have been calculated using previous values.

The Seasonal Naïve model utilised the value from the same workday of the previous week.

The Seasonal Average model was based on the average number of calls received on that workday. This means I looked up the weekday (like Monday, Tuesday, etc) and matched it with its average value. It was simple to apply and ensured consistent results.

The Seasonal Exponential Smoothing model took advantage of Excel's "FORECAST.ETS" function, which accounts for trend and seasonality. Using the "FORECAST.ETS" function, Excel chose the best setting (like alpha). This method saved time and improved model accuracy without needing manual turning (Bruns, n.d.).

Three error measures were calculated to calculate accuracy on the training data (in-sample), and the same error measures were also calculated for the forecasted week (out-of-sample) to check how well each model performed on unseen data (Hyndman & Athanasopoulos, 2021).

- Mean Error (ME) to show bias.
- Mean Absolute Error (MAE) measures the average size of errors.
- Root Mean Squared Error (RMSE) to highlight significant mistakes.

	Seasonal					
Metric	Naïve (In Sample)	Naïve (Out Sample)	Average (In Sample)	Avearage (Out Sample)	Exponential Smoothing (In Sample)	Exponential Smoothing (In Sample)
ME	347.49	9.14	0.00	597.45	44.83	15.64
MAE	478.31	362.29	355.66	597.45	243.80	279.77
RMSE	735.29	464.97	424.16	676.35	308.23	327.69

Figure 5 Seasonal Metrics (ME, MAE, RMSE)

In **Figure 5**, the Exponential smoothing model performed best on the training data, with the lowest in-sample MAE (243.80) and RMSE (308.23). The Seasonal Average model had a perfect in-sample ME (0.00) but performed poorly on test data with high out-of-sample errors. The Naïve model was the least accurate overall in both in-sample and out-of-sample performance. Based on accuracy and flexibility, the Exponential Smoothing model was the best. It adapted to changing trends and gave stable, reliable results for past and future demand.

New Subscription Installation: Inventory Simulation

Inventory system type and key variables:

Delta Sky Broadcasting's local service hubs have a record point inventory system with fixed order quantities. The system keeps a supply of satellite equipment to establish new TV subscriptions. If inventory falls to two or fewer units, a 10-box order is issued to the regional distribution centre. Orders are placed at the end of the day and received at the beginning of the day.

Key variables included:

- Daily installation requests, using a discrete probability distribution
- Boxes per installation, with a discontinuous distribution
- Holding costs £10 per box per day
- Ordering costs £500 per order
- Backorder costs £20 per box per day
- Order quantity (10 boxes)
- Reorder level (<2 boxes)
- Rearranging stock (10 boxes)

This system's advantages include simplicity, ease of implementation, and clear reordering rules. Its disadvantages include the possibility of overstocking (expensive holding expenses) or stockout (penalties for unfilled demand). This system performs well when demand is modest and predictable, but may suffer from excessive variability or unexpected demand spikes.

Simulation Implementation:

The 7-day simulation was generated in Excel, with random numbers derived from the student ID seed. Every day's process is included.

Installation Requests and Demand: Probability tables were used to simulate installation numbers and boxes per installation. For example, on June 26th, one installation required three boxes, resulting in three demand units.

Inventory Balance: The Opening inventory was recorded every morning. After meeting demand and managing backorders, the remaining inventory was estimated before and after making fresh orders.

Orders and Receipts: If the ending inventory was two or fewer, an order of 10 boxes was placed. Orders arrived the next morning. For example, on June 28 and July 1, inventory declined to 1 and 0, respectively, resulting in the new order.

Daily Costs:

Costs are calculated as follows:

- Holding Costs = Closing inventory * £10
- Ordering Costs = £500 if an order is placed
- Backorder Costs = Unfulfilled demand * £20
- Total Cost = sum of the above

Simulation setup and operations:

The simulations relied on explicit assumptions:

- Random numbers were generated using Excel's "Random Number Generator," seeded with the first four digits of the Student ID.
- The demand and installation distributions were simulated using LOOKUP using cumulative probability values.
- Backorders were carried ahead until fulfilled and given priority when new stock arrived.
- Orders are always delivered from their inventory the next day; partial deliveries or delays are unexpected.

Each formula monitored inventory levels, expenses and order timings. The decision included setting a fixed order quantity of 10 and keeping the reorder threshold at two units or fewer. This approach guaranteed timeliness while also managing extra stock.

Total and Average Cost Analysis:

Holding Cost	Ordering Cost	Backorder Cost	Total Daily Costs
70	0	0	70
30	0	0	30
10	500	0	510
90	0	0	90
60	0	0	60
0	500	120	620
80	0	0	80
340	**1000**	**120**	**1460**
Total Costs			
	Average Daily Costs		**208.5714286**

Figure 6 Diff Cost Analysis

The total inventory cost **(Figure 6)** over 7 days was £1460, comprising:

- Holding Costs: £340
- Ordering Costs: £1000 (2 times orders placed)
- Backorder Costs: £120

The average daily cost was £208.57.

There is a clear trade-off:

A higher reorder point or bigger order size might minimise backorders while increasing holding expenses. A lower reorder point increases the likelihood of frequent stockouts, which might result in penalties. This balance must be optimised to save overall costs while maintaining service reliability.

Recommendations:

Delta Sky Broadcasting could improve its inventory system by re-evaluating the reorder point. Increasing it slightly may minimise last-minute shortages and reduce backorders. Consider a variable order quantity: instead of constantly ordering 10 boxes, order only enough to replenish inventories to a certain level. You may predict peak days or seasonal patterns using demand forecasting and place orders ahead. Adding buffer stock: Maintain a modest buffer to absorb unexpected demand increases.

Using this information to refine its inventory strategy, the organisation can enhance installation efficiency, cut wasteful expenditures and deliver better customer service.

Engineer Job Allocation

This section tries to enhance Delta Sky Broadcasting's existing manual approach for assigning installation projects to engineers. The goal is to reduce overall travel distance using a Microsoft Excel linear programming (LP) model. The corporation's current approach yields a total distance of 28 miles. This analysis demonstrates how a more efficient assignment might reduce the distance.

Decision and Decision Variable

Engineer/Customer	Job 1	Job 2	Job 3	Job 4	Job 5
Tom	0	0	1	0	0
Bob	0	0	0	0	0
Paul	0	1	0	1	0
Steve	1	0	0	0	0
Lucas	0	0	0	0	1

Figure 7 Allocated binary values

In this Case **(Figure 7)**, five engineers (Tom, Bob, Paul, Steve, and Lucas) were to be allocated to five customer projects. The goal was to assign each work to only one engineer while minimising overall travel distance. In Excel, binary decision variables were created to represent the assignment choices, with "1" indicating that a specific engineer was assigned to a particular job and "0" indicating that they were not. The variables were placed in a 5x5 matrix for each potential engineer-job combination.

Objective and Objective Function

The goal is to reduce the overall distance travelled by engineers. A table was used to provide distance values between engineers and jobs. The Excel formula for calculating the total journey distance was used: "SUMPRODUCT ()." This formula multiplies the distance matrix by the binary assignment matrix and adds the results, providing the total distance depending on the specified assignments.

Constraints

To ensure an appropriate assignment, the following conditions were imposed

- Each work must be allocated to only one engineer
- Each variable must be either zero or one

The sum function was used in Excel to restrict the allocation of work to each column of the assignment matrix. For example, assume that just one engineer is allocated to job 1. This was repeated for all five positions.

No additional constraints were needed since the problem allowed an engineer to be assigned multiple jobs. The focus was strictly on minimising the distance while ensuring each job is uniquely assigned.

Solver Solution and Optimal Assignment

The problem was solved utilising Excel's Solver and the Simplex LP approach. The goal was to minimise overall distance, and all limits were entered appropriately. The solver successfully identified an optimum solution that met all of the requirements. According to the findings, the best job assignments were Steve to Job 1, Paul to Jobs 2 and 4, Tom to Jobs 3 and 5, and Lucas to Job 5. Bob was not allocated to any jobs, which is understandable because the constraint is that each work must be finished, not every engineer must be employed.

Total Distance and Conclusion

Total Distance	27

Figure 8 Output of Total Distance

The overall distance travelled in this best arrangement was 27 miles (**Figure 8**), instead of 28 miles using the company's existing manual technique. While the difference in this case is only one mile, the advantages of utilising a linear programming method become more apparent when expanding to numerous hubs and hundreds of daily operations. A constant reduction in travel distance may result in cheaper fuel expenses, less vehicle wear, and improved time management for the engineering team.

Linear programming is a more efficient and objective method of assigning engineers to assignments. The organisation may save operational expenses by reducing overall trip distance while improving service delivery. Delta Sky Broadcasting is proposing to implement this concept throughout its service centres. Teams without sophisticated programming expertise may successfully design and maintain this system using simple Excel functions and Solver. Integrating this LP-based assignment mechanism into regular operations would result in apparent cost and productivity improvement.

Engineer Vehicle Servicing and Maintenance

Delta Sky Broadcasting has a fleet of vehicles engineers use for satellite box installation. These vehicles are serviced every Saturday at the local service hub to guarantee continuous efficiency. A simulation was run in Microsoft Excel to assess the efficacy of the present maintenance system. The simulation represented the arrival and servicing of 15 automobiles by two mechanics, with random arrival and service times based on Poisson and exponential distributions, respectively. The student ID’s first four numbers were used as a random seed to ensure consistency and repeatability.

Arrival timings for the 15 cars were simulated using interarrival times from the Poisson distribution, assuming four vehicles arrive each hour. Each interarrival time was estimated using a random integer and then combined to determine each vehicle's real arrival time. Simultaneously, service durations were calculated using an exponential distribution with a mean of 20 minutes to match real-world variation in car servicing time. Each car was given to a mechanic depending on availability, with the one who finished first taking the next vehicle in the queue.

Total Wait Time Metrics	
Total Wait Time	208.27
Average Wait Time	13.88
Maximum Wait Time	41.73

Figure 9 Wait Time Metrics

The Excel model kept account of each vehicle's servicing start, end, and waiting times. If a mechanic was available when the car arrived, it was serviced promptly, with no wait time. If not, the car waited until the allocated mechanic was available. For example, Vehicle 1 was serviced immediately upon arrival. However, Vehicle 2 required the longest wait time of roughly 46.38 minutes before Mechanic 2 could begin. The simulation **(Figure 9)** provided a total wait time of 208.27 minutes, with an average of 13.88 minutes per car. The most excellent wait time reported was 41.73 minutes, indicating congestion during peak hours.

Total Mechanic Idle Metrics	
Total Idle Time for Mechanic 1	129.53
Total Idle Time for Mechanic 2	103.83
Total Simulation Time	317.78
Peak Time for Mechanic 1	180.79
Peak Time for Mechanic 2	213.94
Idle Time Percentage for Mechanic 1	0.41
Idle Time Percentage for Mechanic 2	0.33

Figure 10 Mechanic Idle Metrics

According to the **Figure 10**, Idle time for each mechanic was also measured by comparing the finish time of one work to the beginning of the next. Mechanic 1 had a total idle time of 129.53 minutes, whereas Mechanic 2 had 103.83. Given a total simulation duration of 317.78 minutes, Mechanic 1's idle time percentage was 41%, and Mechanic 2's was 33%. This shows that, while both mechanisms were used relatively successfully, there is still space for improvement in scheduling efficiency to minimise idle time.

The simulation setup calculated important operational KPIS using built-in Excel formulae and logic. Arrival and servicing times were assigned random values, while mechanic availability determined vehicle routing. The model assumed that service is provided on a first-come, first-served basis and that there are no disruptions in the servicing process. Furthermore, each car is allocated to a single technician, and all services are finished on the same day, with no rescheduling.

The findings show that while the existing two-mechanic arrangement can handle the service load quite well, wait times for particular cars can still be high. During peak hours, the gap between arrivals and mechanic availability leads to longer service waits. This influences overall efficiency and may cause engineers to miss deadlines for scheduled projects.

Based on the simulation results, various recommendations might be made. A planned appointment system might help disperse demand more equally throughout the day, lowering

wait time. Second, employing a third technician during peak hours or on busy Saturdays might drastically cut wait times and increase customer satisfaction. Finally, service requests might be prioritised according to vehicle usage frequency, allowing higher-priority cars to be serviced first and reducing operating disturbances.

Finally, the simulation provided valuable insights about mechanic performance, wait times, and idle periods. Using Excel for this queueing simulation revealed critical inefficiencies and places for improvement. Delta Sky Broadcasting may improve the dependability and efficiency of its vehicle service process by making minor modifications to scheduling and resource allocation.

Conclusion

The report analyses Delta Sky Broadcasting's core operating areas to aid decision-making and resource optimisation. The study of new subscription call demand demonstrated strong weekly seasonality, with greater call volumes midweek and reduced weekend activity. A 7-day centred moving average and a seasonal profile chart helped visualise these trends. Among the studied models for estimating future demand, exponential smoothing utilising Excel's FORECAST.ETS function was determined to be the most accurate and flexible.

Inventory simulation over seven days revealed the optimal balance of holding and backorder prices. The current ordering point of two boxes proved to be mostly effective, while backorders did occur, costing £1460 in total and £208.57 per day on average. The engineer's work assignment problem was addressed by linear programming, lowering the overall trip distance from 28 to 27 miles. Finally, a queueing simulation of car service revealed modest wait times and notable idle intervals for mechanics, indicating that scheduling may be improved.

It recommends that the company significantly boost its reorder point to prevent stockouts, regularly use LP-based engineer assignments, and consider adding scheduled servicing or a third mechanic during high demand to improve operational efficiency.

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